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Accelerating Innovation Culture: A Data-Driven Transformation Journey Across Complex Organizations

T. Schroeckenfuchs, OMV E&P GmbH, Vienna, Austria; F. Wedam and M. M. Voda, OMV Petrom, Bucharest, Romania; A. Glaser, OMV Downstream GmbH, Vienna, Austria; J. Rust, Borealis AG, Stenungsund, Sweden; B. Campbell, OMV E&P GmbH, Vienna, Austria; A. Leitner, OMV AG, Vienna, Austria

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Abstract

This paper aims to explore how a large, integrated multinational group of organizations can systematically assess and enhance its innovation culture. The objective is to identify key barriers and enablers of innovation using data-driven methods, and to propose a scalable framework that supports cultural transformation across complex, multi-disciplinary and diverse environments. A strong innovation culture is positioned as a critical enabler for successful transformation in the energy and petrochemical industries. A comprehensive survey was designed to capture both qualitative and quantitative insights into the current state of innovation culture. The survey included open-ended, ranked, and scaled questions, and was distributed across all sub-companies of the corporate group, encompassing a wide range of business units and seniority levels. Data calibration ensured demographic representativeness. Analytical methods included exploratory data analysis, Total Unduplicated Reach and Frequency (TURF) analysis, and Performance-Importance Matrices (PIMs). Open-ended responses were thematically coded to extract conceptual insights. The findings informed a strategic roadmap supported by inclusive initiatives such as innovation weeks, hackathons, and continuous learning platforms. The analysis grouped innovation barriers into three overarching categories - leadership (e.g., lack of resources, inconsistent support), organizational (e.g., unclear accountability, inefficient decision-making), and practical (e.g., fear of failure, weak collaboration) - drawing on established academic frameworks for understanding innovation challenges. The most significant barrier was fear of failure which if lowered in the organization would have the highest impact and greatest reach and therefore identified as a high priority improvement area. Key enablers included leadership support, acceptance of failure, resource allocation, and cross-functional collaboration. These insights led to the development of a multi-year implementation plan emphasizing inclusivity, targeted problem-solving, and agility. The approach demonstrates how data science can be leveraged to diagnose cultural dynamics and prioritize interventions that foster innovation. The framework is adaptable and can be applied across industries seeking to embed innovation as a core organizational capability. This paper introduces a replicable, data-driven methodology for diagnosing and enhancing innovation culture. It bridges employees and groups development with data

science, offering actionable insights and scalable strategies that can benefit transformation efforts across the petrochemical and energy sectors.

Introduction

Innovation is increasingly recognized as a strategic capability in the energy and petrochemicals sector, enabling organizations to respond to the growing complexity of global energy systems, climate imperatives, and technological disruption (e.g. [Costa-Campi, 2014](#), [OECD 2023](#)). The International Energy Agency (IEA, 2025) emphasizes that sustained innovation is essential to meet climate and energy security goals, while [Popp et al. \(2020\)](#) highlight that structural changes in energy markets demand continuous innovation and entrepreneurship.

Empirical research confirms that innovation capability plays a central role in translating dynamic capabilities into competitive advantage and performance ([Ferreira et al., 2020](#)). Studies across various sectors show that innovation culture acts as a catalyst for creativity and problem-solving, ultimately influencing a company's performance and adaptability ([Kharisma et al., 2025](#); [Bayhan and Korkmaz, 2021](#); [Uzkurt et al., 2013](#); [Hazem and Zehou, 2019](#)), especially when organizations encourage fresh thinking and challenge conventional assumptions ([Bouquet et al., 2021a](#), [Bouquet et al., 2018](#)). This reinforces the view that innovation is not merely a cultural aspiration but a measurable and actionable driver of strategic company success. Innovation is also widely acknowledged as a core corporate value appearing in 80% of Standard & Poor's 500 firms ([Guiso et al., 2015](#)), once more, reinforcing the above-mentioned benefits.

As the industry transitions toward more sustainable and efficient operations, cultivating a robust innovation culture becomes critical, not only to remain competitive but to ensure long-term resilience and relevance.

Despite its prominence, many frameworks associated with innovation and organizational culture, such as the ETOILE model ([Davies & Buisine, 2018](#)), the Competing Values Framework ([Cameron & Quinn, 2011](#)), or absorptive capacity theory ([Zahra & George, 2002](#)), remain largely conceptual and idealized. While they offer valuable perspectives on organizational dynamics, they are rarely grounded in direct feedback from employees or tailored to the specific challenges of individual industries. Yet it is precisely these lived experiences that should inform the design of environments and initiatives (e.g. [Birkinshaw et al., 2011](#)) aimed at fostering a meaningful and effective innovation culture.

While empirical studies have begun to explore innovation culture in various sectors and SMEs (e.g. [Latif et al., 2025](#), [Alkhalaf et al., 2025](#)), there remains a notable gap in research from the energy and petrochemical industries. Few studies examine how innovation is perceived by the workforce or identify the barriers employees face in practice (e.g. [Sharma et al., 2025](#), [Roberts et al., 2023](#)).

To address this gap, the present study investigates how innovation is perceived and experienced by employees in a cross-section of the energy and petrochemical sector. Drawing on selected, structured interviews conducted across the three multinational companies OMV, Borealis, and OMV Petrom, the research captures insights from various organizational levels and business areas. Conducted as part of a cross-company leadership development initiative, the study represents a practitioner-led pilot designed to test a structured, bottom-up methodology for assessing innovation culture. Unlike many existing studies that focus primarily on leadership perspectives, this approach emphasizes the lived experiences of the workforce and the systemic barriers they encounter.

Building on this empirical foundation, the paper proposes a scalable framework for cultural transformation that embeds innovation into everyday work, not only in technical domains, but across the entire organizational ecosystem. By aligning cultural enablers with operational realities, the framework aims to foster creativity, support collaboration, and create space for new ideas to grow, ensuring that the organizations remain agile, resilient, and future-ready in a rapidly evolving energy landscape.

While the proposed framework is tailored to the specific context of the participating companies, the structured interviews and subsequent data analysis approach used in this study demonstrates its potential as a flexible diagnostic tool. It offers a fast and accessible way to assess innovation culture, helping organizations identify the barriers and enablers that shape their innovation efforts. It provides a clear and actionable starting point for targeted cultural interventions and long-term development.

Methodology

Overview and Theoretical Framing

This study applied a mixed-methods approach, combining structured interviews, survey data, and advanced data science techniques, to explore innovation culture within a multinational group of companies in the energy and petrochemical sectors. The objective was to develop a practical, scalable framework for fostering innovation culture based on empirical insights.

Innovation culture is defined here as the shared mindset, behaviors, and systems that support the generation and implementation of new ideas across all levels of the organization. This aligns with [Schein's \(1992\)](#) view of organizational culture as learned patterns of assumptions that guide behavior. Innovation culture in this context includes not only technological advances but also process improvements, customer-centric solutions, and sustainability-driven initiatives.

The study draws on established academic frameworks in innovation management theory and organizational behavior (e.g., [Tidd et al. 2005](#), [Bouquet et al. 2021b](#), [Osterwalder et al. 2020](#)), which informed the design of the survey and interview instruments. The applied tools were developed to capture both qualitative and quantitative insights into how innovation is perceived and practiced across the organization.

Research Design and Data Collection

A comprehensive survey was conducted across an integrated multinational group of companies operating in oil and gas, low-carbon energy, and petrochemicals. It targeted a wide range of business units, seniority levels and countries, aiming to generate actionable insights and identify barriers to innovation.

Survey Structure. The survey was designed to capture both qualitative and quantitative data through a mix of question formats:

- **Open-Ended Questions** allowed respondents to share personal experiences and perspectives, providing rich qualitative input.
- **Top Three Selections** helped prioritize key issues by identifying the most critical factors.
- **Scaled Questions** (e.g., Likert scales) enabled the measurement of attitudes and perceptions, supporting statistical analysis.

This multi-format approach provided a holistic view of how innovation is perceived and practiced across the organization. Follow-up interviews were conducted to explore the nuances behind survey responses.

Sampling and Distribution. The survey was distributed across different levels of seniority and departments within the group of organizations, ranging from field technicians and operational staff to senior management (EB-1 level). Such coverage ensured that perspectives from all organizational layers were captured, enhancing the validity and strategic relevance of the insights. A total of 60 responses were collected. To correct for sampling biases, a weighting process was applied based on company affiliation and gender, ensuring demographic representativeness.

Limitations and Scope. It is important to note that the survey was conducted as part of a cross-company leadership development program, designed to explore how innovation culture could be strengthened across

companies, locations and business areas. It was not a top-down, centrally designed capability-building initiative, but a practitioner-led pilot developed and executed by a gender-balanced, multinational team of six mid- to senior-level managers from diverse business areas. The approach prioritized depth over breadth: participants provided written feedback and engaged in face-to-face interviews, which allowed them to reflect meaningfully on the topic and clarify the interpretation of specific innovation barriers. Rather than aiming for comprehensive employee feedback, the study was intended to test the value of a structured, bottom-up, data-driven methodology. While responses were received from multiple countries, the sample size (60) remains the main limitation. Future research should build on this foundation by expanding the dataset and validating the framework across broader organizational and cultural contexts.

Data analyses Approach

Application of Data Science in Survey Analysis. To extract meaningful insights from the survey data, a structured data science approach was applied. This included techniques from statistics and exploratory data analysis, enabling the identification of patterns, trends, and correlations across organizational levels. Rather than relying solely on descriptive statistics, the analysis integrated both structured (e.g., scaled responses) and unstructured (e.g., open-ended responses) data. This allowed for a comprehensive understanding of how innovation is perceived and practiced across the organization.

Analytical Process. Figure 1 outlines the complete analytical process followed in this study, from initial data preparation through to exploratory and advanced analytical techniques.

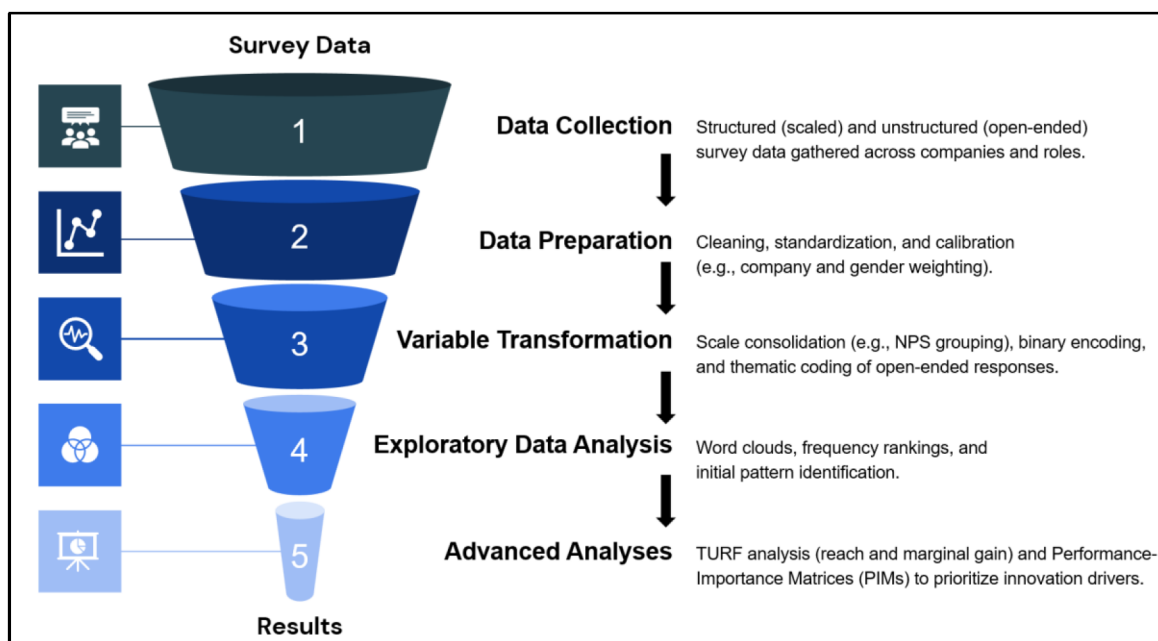


Figure 1—Analytical workflow for the conducted innovation survey evaluation.

Data Preparation. Before analysis, the dataset was cleaned and transformed to ensure consistency and analytical validity. Calibration techniques were applied to correct for sampling biases and ensure demographic representativeness.

- **Sample Calibration:** Responses were weighted by company and gender to accurately reflect the organizational population.
- **Variable Treatment:** Several transformations were performed to optimize the signal-to-noise ratio in the data:

- **Scale Consolidation:** Likert-scale responses were grouped using Net Promoter Score (NPS) methodology to distinguish promoters, passives, and detractors.
- **Binary Encoding:** Categorical selections were converted into binary variables to simplify interpretation.
- **Conceptualization of Open-Ended Responses:** Open-ended responses were thematically coded into key concepts, facilitating thematic analysis and identification of common themes.

Exploratory Data Analysis (EDA). Exploratory data analysis is a critical step in understanding the data and identifying initial patterns and trends. The initial analysis focused on identifying dominant themes and stakeholder priorities.

- **Word Clouds:** Visual representations of the frequency of words in open-ended responses highlighted core themes and concepts.
- **Item Rankings:** Survey items were ranked based on their frequency and importance, providing insights into the most critical factors according to stakeholders.

Advanced Analytical Techniques. To deepen the insights and enable the identification of focus areas for initiatives with the highest reach and frequency, two advanced methods were applied:

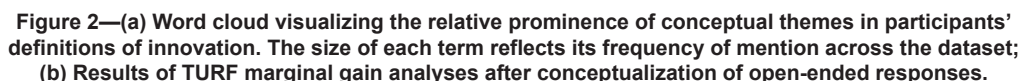
- **Total Unduplicated Reach and Frequency (TURF) Analysis.** TURF analysis was used to identify barriers and actions with the highest reach and frequency. This technique helped in understanding the extent to which different factors impact the innovation culture and in prioritizing the most impactful measures.
 - **Reach:** The proportion of respondents who identified a particular barrier or action.
 - **Frequency:** The number of times a particular barrier or action was mentioned by respondents.
- **Performance-Importance Matrixes (PIMs).** Performance-Importance Matrices were deployed to assess the importance and performance of various aspects related to innovation culture perception. This involved plotting factors on a matrix to identify areas where performance is high, but importance is low, and vice versa.
 - **Importance:** The significance of a factor in influencing innovation culture.
 - **Performance:** The current state of a factor within the organization.

Presentation of Data and Results

Innovation Perception

Participants were asked to describe what innovation means to them in their own words. The responses demonstrated a rich and multifaceted understanding of innovation across the Group, with no two answers being identical. [Figure 2a](#) presents a visual representation of these results in the form of a word cloud.

Several dominant concepts emerged. Many respondents emphasized innovation as a departure from established routines, challenging the status quo and rethinking how work is done (Novelty). Innovation was frequently associated with improving processes, reducing effort, and increasing operational efficiency, optimization and effectiveness. Participants linked innovation to tangible business outcomes and value creation, such as cost reduction, enhanced customer satisfaction, and competitive advantage. Numerous responses referenced the role of emerging technologies, digital tools, AI and data-driven methods in driving their perception of innovation. Innovation was also described as a mindset requiring openness to change, curiosity, and a willingness to experiment and learn from failure. Most respondents highlighted the



After conceptualizing the open-ended responses, a TURF analysis was conducted (Fig. 2b) to quantify how widely certain concepts are mentioned and to identify which combinations of perceptions maximize overall coverage. This approach helped to clarify which themes are most impactful for targeted action. Concepts such as "efficiency" and "doing things differently" emerged as particularly valuable. Although they were mentioned less frequently, they contributed significantly to overall reach, with high marginal improvements, indicating that they capture unique perspectives not covered by other categories. In contrast, more commonly mentioned concepts like "novelty" and "challenging the status quo" showed lower marginal contributions, suggesting that while they are widely recognized, they offer limited additional reach when included in a conceptual model.

Responses to the open-ended question "In what ways could innovation drive growth in your business area?" revealed a broad and strategic understanding of innovation as a growth driver (Figure 3a).

Many participants emphasized automation, process optimization, and resource savings as pathways to growth. Innovation was linked to the use of advanced technologies such as AI, digital twins, and machine learning to improve decision-making, efficiency and operational speed. Respondents highlighted the role of innovation in developing low-carbon solutions, complying with environmental regulations, accessing new markets and delivering on market needs (e.g., CCS, low carbon hydrogen). Innovation was seen as a means to rethink traditional approaches, create new revenue streams and new business opportunities to maintain competitiveness. Several responses pointed to the importance of innovation in empowering teams, improving collaboration, and enhancing visibility and strategic alignment.

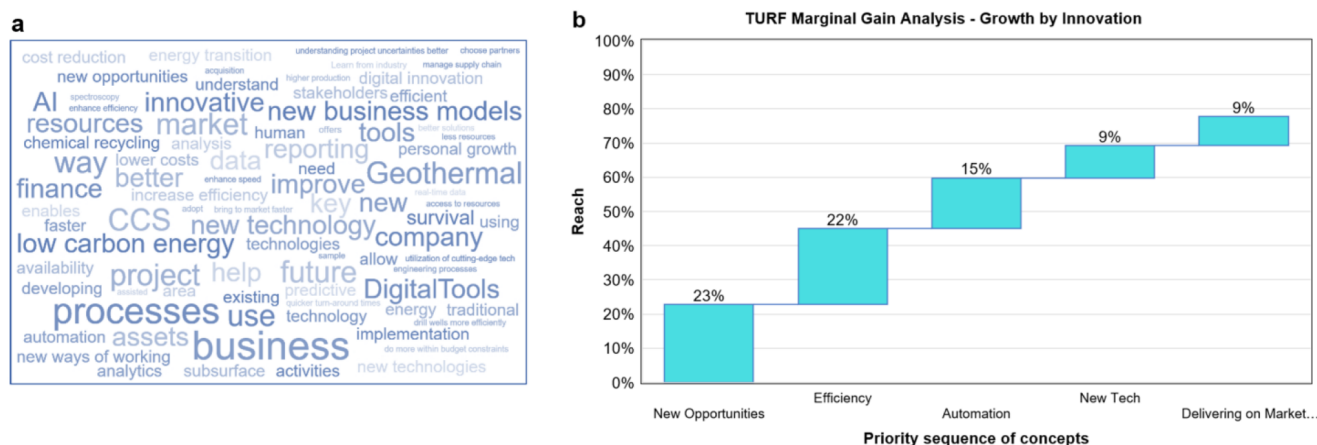


Figure 3b illustrates how participants perceive innovation as a driver of business growth, based on a TURF analysis. Concepts such as "efficiency" and "new opportunities" stand out for their high marginal contributions, despite being mentioned less frequently than other themes. Meanwhile, more frequently mentioned ideas like "automation", "new technology", and "delivering on market needs" show lower marginal improvements, suggesting that while they are widely recognized, they offer less incremental reach. These findings point to a pragmatic orientation toward innovation, with a strong emphasis on operational and technological drivers.

Innovation-Level Assessment

As part of the study, participants were asked to respond to the question: "How innovative would you rate your business area?" Responses were given on a numeric Likert-type scale from 0 (not at all innovative) to 10 (extremely innovative), reflecting both internal and external perceptions of innovation within their respective roles and contexts.

The overall average score across all responses was 5.88, suggesting that innovation is present but not yet perceived as a core strength in many areas. The distribution of responses clustered primarily in the mid-range (scores 5–7), indicating cautious optimism. Only one respondent rated their area as fully innovative (score 10), while a few rated it very low (scores 2–3), pointing to perceived gaps in innovation maturity perception (Fig. 4a).

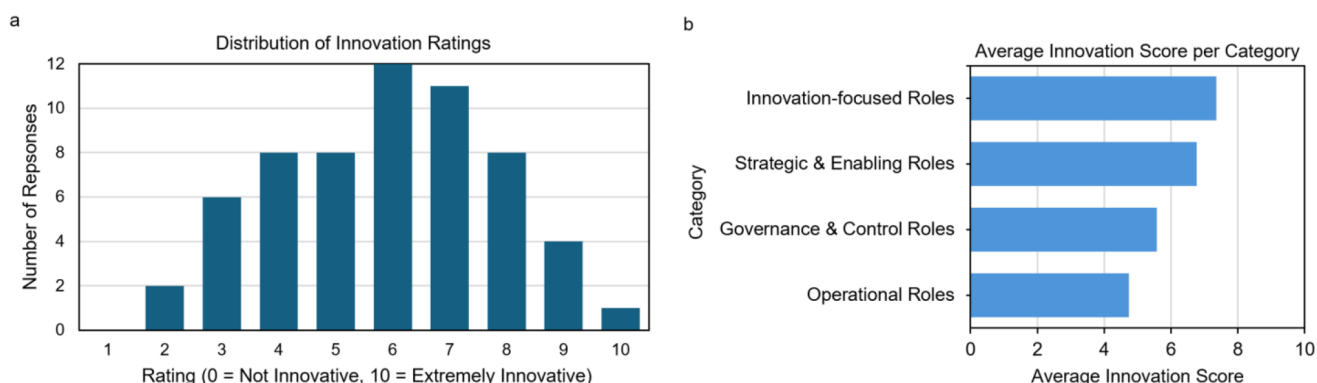


Figure 4—Distribution of Innovation rating; (a) Overall distribution and (b) Average Innovation Score per Category showing that innovation is most strongly perceived in roles explicitly designed to foster it.

To gain deeper insight, responses were grouped into 4 functional categories (Fig. 4b): Innovation-focused roles (e.g., Technology & Innovation, Low Carbon Business, Circular Economy) received the highest average score of 7.36, reflecting strong alignment with innovation objectives and visibility. These roles are typically tasked with driving change and exploring new solutions, which may explain their more favorable self-assessment. Strategic & Enabling roles (e.g., Strategy, Business Transformation, IT) showed a moderate to high average score of 6.77. These functions often support innovation through strategic direction, stakeholder engagement, and transformation initiatives. Governance & Control roles (e.g., Finance, Controlling, Compliance) had an average score of 5.57. While not directly responsible for innovation, these areas may support or constrain it through policy, risk management, and resource allocation. Operational roles (e.g., HSSE, Refinery Site Management, Operations, Supply Chain) had the lowest average score of 4.73. These areas are typically focused on business continuity, safety, and reliability, which may reduce flexibility and risk tolerance and therefore perception of innovation.

This functional breakdown highlights that innovation is most strongly perceived in roles explicitly designed to foster it, while more traditional or efficiency-driven functions may require targeted support to cultivate innovative thinking.

Barriers to Innovation

To complement the perception-based insights, the next part of the analysis focused on identifying the structural and behavioral factors that either hinder or support innovation within the organization. This section drew on both empirical data and established academic frameworks to categorize and evaluate the most critical barriers ("headwinds") to innovation.

The analysis grouped innovation barriers into three overarching categories: Leadership Support, Organizational Design, and Innovation Practice, based on the framework introduced by Osterwalder et al. (2020). These categories were further refined using practitioner input, including a set of unpublished innovation blocker cards shared by Cyril Bouquet, which detail the specific barriers outlined below:

- **Leadership barriers** (lack of recognition and rewards, neglecting development, overemphasis on hierarchical decisions, lack of inclusivity, inconsistent messaging, blame culture, short-termism, micromanagement, resistance to change, ignoring employee input, inconsistent support, lack of vision, insufficient time for innovation, lack of resources, failure to learn from mistakes)
- **Organizational barriers** (unclear accountability, overemphasis on compliance, inadequate reporting and feedback loops, fragmented communication tools, inadequate technology infrastructure, unclear roles and responsibilities, resource constraints, limited transparency in resource allocation, inadequate innovation metrics for evaluating success, rigid project management, non-transparent decision making, resource allocation barriers, inefficient decision-making structures, rigid hierarchy)
- **Practical barriers** (inadequate skill development, lack of diversity and inclusion, reluctance to kill flawed projects, inadequate innovation ecosystem (partnerships and external view), inconsistent evaluation criteria, ad hoc idea handling (ideas collected informally and inconsistently), little encouragement to learn from mistakes, weak cross functional collaboration, lack of agile methodologies, reluctant to experiment – fear of failure, insular thinking, inadequate prototyping and testing, weak customer engagement)

For each category three key barriers to innovation were identified, categorized by their prevalence and marginal impact on innovation culture. Marginal impact refers to the additional proportion of respondents affected when a specific barrier is added to the set of already considered concerns, helping to prioritize those with the greatest influence on perceptions. To ensure broad organizational relevance, the selected combinations of barriers within each category collectively account for approximately 80% of respondent concerns. A visual summary of these findings is presented in Fig. 5.



Figure 5—Barriers to innovation that concern around 80% of the respondents in the three categories leadership support, organization design and innovation practice.

The following barriers were identified:

1. Leadership Barriers:

- **Lack of Resources:** The most prominent leadership-related barrier, with a marginal impact score of 0.450, was the limited allocation of key resources such as time, budget, and personnel for innovation activities. The lack of adequate resources hampers the ability to pursue innovative projects and initiatives.
- **Inconsistent Support:** Inconsistent support from leadership was another critical barrier, with a marginal impact score of 0.188. This inconsistency potentially leads to uncertainty and reluctance among employees to engage in innovative activities.
- **Hierarchical Decisions:** A tendency toward hierarchical decision-making, where input from others may be limited or overridden, was identified as a barrier with an impact score of 0.141. Such approaches can stifle creativity and slow down the implementation of innovative ideas.

2. Organizational Barriers:

- **Unclear Accountability:** A lack of clarity regarding roles and responsibilities in innovation processes was identified as a key organizational barrier, with a marginal impact score of 0.395. This ambiguity can lead to inefficiencies and hinder the effective execution of innovative initiatives.
- **Inefficient Decision-Making:** Slow or overly complex decision-making structures were noted as a challenge with impact score: 0.247. These processes can delay approvals and reduce the agility needed to advance innovative ideas.
- **Resource Constraints:** Resource constraints were identified with a marginal impact score of 0.216, potentially limiting the ability to allocate the necessary support such as time, budget, or personnel for innovation activities.

3. Practical Barriers:

- **Fear of Failure:** Fear of failure was the most significant practical barrier, with a marginal impact score of 0.552. This fear discourages employees from taking risks and experimenting with new ideas.
- **Weak Collaboration:** Weak collaboration among teams, with a marginal impact score of 0.226, prevents the sharing of knowledge and ideas, which is essential for innovation.
- **Lack of Agile Methods:** The lack of agile methods, with a marginal impact score of 0.088, slows down the development and implementation of innovative solutions.

To integrate perceptions of innovation barriers with the overall state of innovation culture, Performance-Importance Matrices (PIMs) were applied (Fig. 6). These matrices reveal the relationship between how detrimental a specific barrier is perceived to be and how frequently it is reported across the organization. Each quadrant of the matrix carries distinct implications. Of particular interest is the upper-right quadrant, which highlights factors rated as both highly important and underperforming—indicating areas that may require immediate attention.



Figure 6—Performance-Importance Matrices (PIMs) summarizing the most critical barriers across the three categories. The upper-right quadrant highlights factors rated as both highly important and underperforming, indicating priority areas for targeted improvement.

The PIMs analysis revealed distinct priority areas within each barrier category. For leadership-related barriers, the most critical issues—positioned in the upper-right quadrant—were identified as lack of resources, insufficient time for innovation, and short-termism, indicating systemic constraints in strategic support. Within the organizational category, top concerns included resource constraints, unclear accountability, and inefficient decision-making structures, reflecting structural misalignments that hinder innovation flow. For practical barriers, the most impactful factors were fear of failure, an inadequate innovation ecosystem, and weak cross-functional collaboration, pointing to cultural and operational challenges at the execution level.

Taken together, the results highlight a consistent pattern across categories: resource limitations, unclear structures, and cultural inhibitors such as fear of failure emerge as recurring themes. This alignment suggests that innovation barriers are not isolated to specific domains but rather reflect systemic challenges that span leadership, organizational design, and day-to-day practices. Addressing these interconnected issues may therefore require cross-functional interventions and a coordinated effort to strengthen both strategic commitment and operational agility.

Enablers to Innovation

Participants were asked to describe organizational structures, procedures, and cultural shifts that could enhance innovation. The responses revealed a diverse set of enablers, many of which align closely with the barriers previously identified.

A word cloud and a TURF analysis summarizing the most frequently mentioned enablers and actions is shown in Fig. 7.

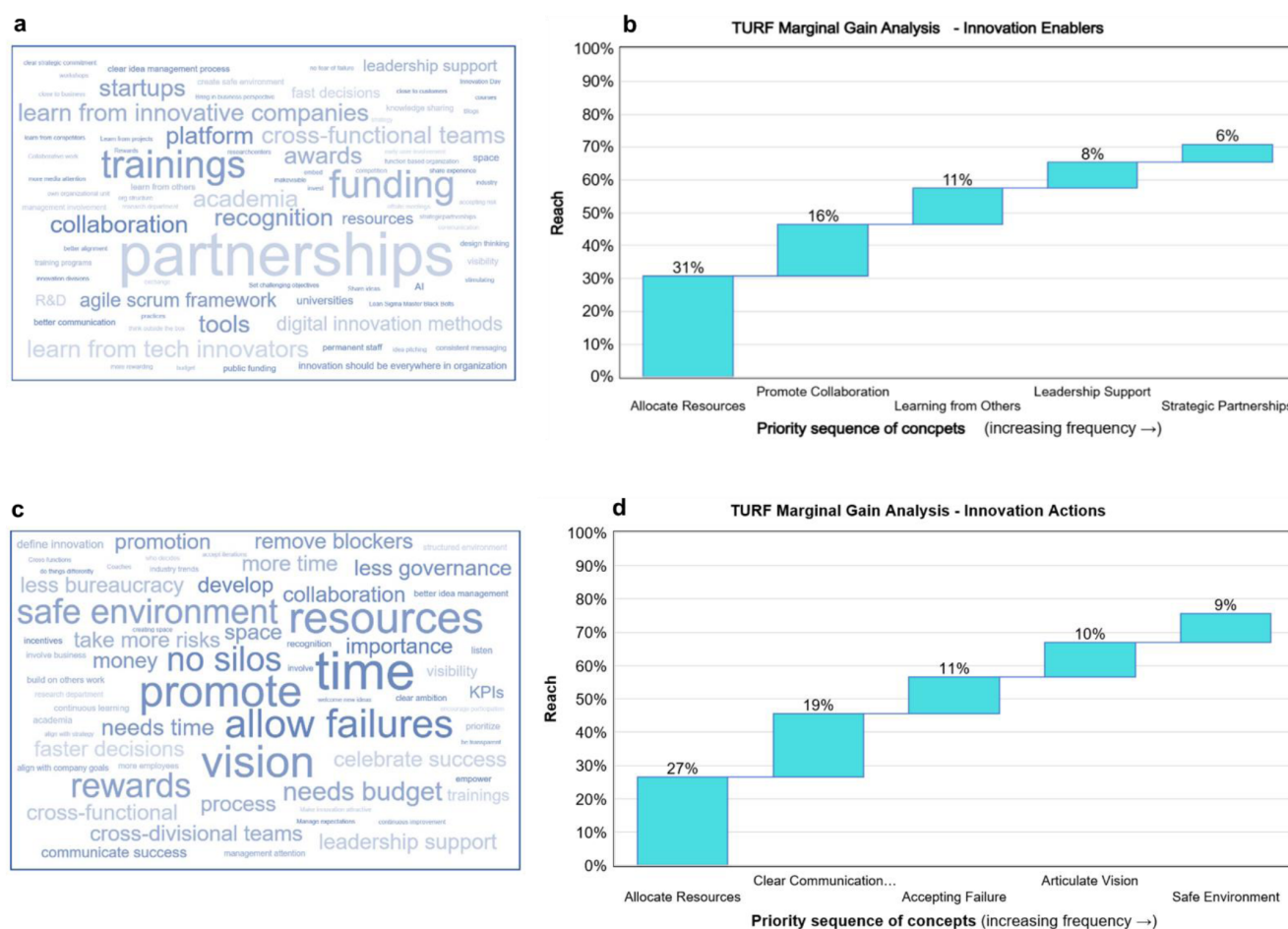


Figure 7—Word clouds visualizing the relative prominence of conceptual themes in participants' definitions of (a) "Innovation Enablers" and (c) "Actions to improve Innovation Culture"; the size of each term reflects its frequency of mention across the dataset. (b, d) Results of the respective TURF marginal gain analyses after conceptualization of open-ended responses.

Several dominant themes emerged. Many respondents emphasized the need for consistent leadership support, clear strategic vision, and adequate resource allocation—highlighting the foundational role of leadership in fostering innovation. Cultural aspects such as accepting failure, encouraging risk-taking, and creating a safe environment were frequently cited as essential for enabling experimentation and learning. Organizational enablers included promoting collaboration, establishing strategic partnerships, and improving cross-functional communication, all of which support knowledge sharing and agility. Respondents also stressed the importance of clear communication of success stories to build momentum and reinforce innovation as a shared goal. While a few respondents view innovation as the responsibility of dedicated research or innovation departments, the majority believe it should be embedded throughout the entire organization.

Together, these insights point to a multifaceted understanding of what it takes to strengthen innovation culture—spanning leadership behavior, organizational design, and everyday practices.

Discussion and Proposed Strategies

Innovation Perception and Strategic Implications

The evaluation of innovation perception revealed a lack of shared understanding across the participating organizations. While the diversity of interpretations, ranging from novelty and digitalization to efficiency and mindset, reflects a healthy breadth of perspectives, it also underscores the need for a unified definition

and strategic vision. Innovation should not be confined to R&D and technology functions; rather, it must be recognized as a core capability across all business areas, essential for enabling growth and transformation.

Consistent with the growth potential assessment, respondents demonstrated a pragmatic view of innovation. It is seen not merely as ideation or basic research, but as a business-critical skill, a lever for efficiency, tangible outcomes, and long-term competitiveness. Innovation was frequently described as a "survival skill" in the context of industry transformation.

The growth potential analysis identified key focus areas where innovation is expected to deliver the highest impact. These areas can serve as strategic guideposts for shaping the innovation framework and prioritizing initiatives:

- Automation
- Efficiency gains
- Deployment of advanced technologies (e.g., AI, digital twins)
- Low-carbon solutions
- Market expansion

Innovation Maturity and Capability Gaps

The average self-assessed innovation maturity score was 5.88 out of 10, indicating moderate development and a good potential for growth. Marked disparities were observed between innovation-focused roles, which reported the highest self-assessed innovation scores (7.36), and operational roles, which rated themselves significantly lower (4.73). This gap appears to be closely linked to the identified barriers, including limited time and resources for innovation, as well as a prevailing business emphasis on rapid delivery and short-term results, conditions that leave little room for experimentation or failure. It highlights the need to embed innovation capabilities more broadly, beyond dedicated innovation teams, and to target support where maturity is lowest. The innovation score could also serve as a KPI to monitor progress and evaluate the effectiveness of future interventions.

Barrier Analysis and Priority Areas

Across the three predefined categories, the most critical barriers based on combined marginal impact (80%) and PIMs analysis—were:

- Leadership: Lack of resources, inconsistent support, hierarchical decision-making, insufficient time, and short-termism
- Organizational: Unclear accountability, inefficient decision-making, and resource constraints
- Practical: Fear of failure, weak collaboration, lack of agile methods, and inadequate innovation ecosystems

Among these, fear of failure emerged as the most pressing barrier, with the highest Level of Association (0.55) and Negative Importance (0.19) in the PIMs analysis. This indicates a strong correlation with poor innovation outcomes and a significant detrimental impact. Addressing this barrier should be a strategic priority, particularly through initiatives that foster psychological safety, experimentation, and cultural resilience.

Enablers and Strategic Levers

For the evaluation of enablers, participants emphasized the importance of aligning support mechanisms with the previously identified barriers. Beyond this alignment, several additional insights emerged. These included the need for clear communication of success stories to foster engagement and visibility, the value

of partnerships and learning from external sources to broaden perspectives, and the motivational impact of recognition and awards. Furthermore, there was a repeated call for a shared definition and strategic vision for innovation, reinforcing earlier findings from the innovation perception analysis.

Integrated findings

The integrated insights from the innovation perception analysis, maturity assessment, barrier identification, and enabler evaluation collectively form key learnings that build the foundation of the proposed innovation framework. These learnings are synthesized into three core dimensions, each directly reflecting the empirical findings from the conducted analysis and addressing the cultural, structural, and operational factors influencing innovation within the organization:

1. **Be Inclusive:** This dimension calls for making innovation accessible to all roles and departments, rather than limiting it to specialized teams. It should foster broad engagement and ensure that innovation becomes a shared organizational capability. It must address barriers such as insufficient time, hierarchical decision-making, lack of resources, inconsistent leadership support, and fear of failure. Special attention should be given to groups with low innovation self-assessment, and the identified growth potential areas should guide the focus of inclusive innovation efforts.
2. **Be Specific:** This dimension requires the implementation of structured formats and targeted interventions to overcome cultural and practical barriers. It should include mechanisms for experimentation, cross-divisional collaboration, and agile working methods. It is intended to tackle barriers such as weak collaboration, lack of agile practices, and inefficient decision-making, while ensuring that innovation activities lead to tangible and measurable business outcomes.
3. **Provide Fundamentals:** This dimension emphasizes the need for visible leadership, clear ownership, effective communication, and accessible tools. It should create the foundational conditions for sustained innovation performance. Key barriers to address include inadequate innovation ecosystems and unclear accountability. It should provide a clear definition of innovation, a strategic vision, recognition mechanisms, training opportunities, and consistent communication of success stories.

These dimensions translate the empirical findings into actionable focus areas, setting the stage for the Innovation Culture Framework proposed below. While structural changes could further support innovation, they were outside the scope of this study. The focus was on identifying actionable cultural levers that can be activated without major reorganization.

Innovation Culture Framework Suggestion

Figure 8 illustrates how the empirical findings, structured into the dimensions Be Inclusive, Be Specific, and Provide Fundamentals, are translated into the proposed innovation culture framework. The framework offers actionable responses to each dimension, outlining concrete mechanisms and initiatives:

Innovation Culture Framework			
Core Dimensions	Framework Actions	Details	Barriers Overcome
 Be Inclusive involve entire organization	Innovation Week Group-wide hands-on Innovation Challenge	• 1 week per year • full focus on hands-on innovation • reaching every employee • quick wins in own business area • from idea to implementation • thousands of improvements • awards & recognition	• Insufficient time • Hierarchical decision-making • Lack of resources • Fear of failure
 Be Specific solve key strategic challenges	Hackathons Cross-divisional expert rounds to solve key challenges	• cross-divisional teams (+externals) • for most pressing challenges • topics with key strategic focus • held in 3 core locations • winning ideas to be implemented • learn agile methods for every-day	• Weak collaboration • Lack of agile methods • Inefficient decision making
 Provide Fundamentals 365 days of exchange, training & information	Innovation Platform Digital hub for innovation enablement	• facilitate Innovation Week and Hackathons (e.g. AI Coach) • Year-round trainings and events • Communication campaigns • Definition, Strategy & Contacts • Innovation metrics (KPIs)	• Unclear accountability • Inconsistent support • Inadequate innovation ecosystem

Figure 8—Overview of the proposed Innovation Culture Framework

Be Inclusive – Innovation Week. Innovation Week is a structured, seven-day initiative facilitated via the Innovation Platform (described below), designed to engage all employees in hands-on innovation. Participation is mandatory and intentionally disruptive to daily routines, ensuring that innovation becomes a shared responsibility across the organization. Employees form location-based teams, with encouragement for cross-functional and external collaboration, and work on strategic themes derived from identified growth potential areas. The week includes leadership-led kickoffs (opened by Board members), targeted training, team work sessions, and multi-level presentations. While HSSE-critical and operations staff may be excluded due to shift constraints, they are explicitly invited to participate in future cycles to ensure inclusivity and avoid repeated exclusion of those with lower innovation self-assessment. The focus is not on developing new business models but on generating ideas with direct relevance to participants' own business areas, such as automation, efficiency, or simplification. Local judging panels evaluate projects based on impact, feasibility, scalability, and originality, with selected teams presenting to senior leaders for recognition. The initiative aims to generate quick wins and embed innovation practices into daily work.

Be Specific – Innovation Hackathons. To address specific strategic challenges with speed and precision, Hackathons are introduced as focused, high-impact formats that complement the broader Innovation Week. While Innovation Week fosters broad engagement, Hackathons are designed to bring together cross-divisional expertise and external perspectives to solve the most pressing business problems. Held in three core locations to also foster cultural and geographical diversity, they convene selected participants through nomination and application, ensuring diversity in skills, backgrounds, and geographies. Each Hackathon centers on a clearly defined challenge aligned with strategic priorities, with teams working intensively over two days to develop actionable solutions. The format includes ideation, training on agile working methods, concept development, and pitching, supported by experienced facilitators and external experts. Winning ideas are handed over to business owners for fast-track validation and implementation. Beyond delivering results, Hackathons serve as a learning platform, equipping participants with innovation tools, agile methods, and new ways of thinking that they can apply in their daily work. To build a sustainable innovation community, participation is rotated annually to expand the pool of trained and engaged employees, ensuring long-term capability building and organizational reach.

Provide Fundamentals – Innovation Platform. The innovation platform serves as the central digital hub for all innovation-related activities providing the structural foundation needed to embed innovation into the daily rhythm of multiple organizations. It supports the strategic vision of making innovation a core capability by offering clarity on what innovation means in the business context. Designed to overcome barriers such as unclear accountability, inconsistent support, and limited access to tools, the platform enables seamless participation in Innovation Week and Hackathons while supporting year-round engagement. It features a modular interface with spaces for event registration, challenge briefings, team collaboration, and implementation tracking. To enhance engagement, gamification elements allow employees to monitor their personal innovation score in an interactive way, linked to the group's value framework. The platform also connects employees with relevant learning content and peer communities and uses skill-based matching to link individuals with innovation-relevant expertise to suitable challenges. An integrated AI-assisted innovation coach guides users through the ideation process, helping refine ideas and navigate available resources. Crucially, the platform is fully integrated with the group-wide idea management system, enabling structured campaigns, topic sourcing, voting, and transparent tracking of idea implementation. By centralizing tools, communication, and progress monitoring—including real-time visibility of KPIs—the innovation platform ensures transparency, continuity, and alignment with business priorities. Over time, it is designed to grow into a learning and collaboration ecosystem—building innovation capabilities across the organization and expanding the community of trained, engaged employees who bring innovation tools and mindsets into their everyday work.

The implementation of the innovation culture framework is structured as a long-term roadmap, beginning with a focused preparation phase ahead of the first Innovation Week. This includes leadership alignment, organizational readiness, and the launch of the innovation platform as the central digital enabler. Governance and ownership of the framework are anchored within the Innovation & Technology team, with cross-functional input to ensure relevance, continuity, and alignment with business priorities. Innovation Week, Hackathons, and the innovation platform form the core pillars of engagement, each supported by a practical monitoring framework. Key performance indicators—such as participation rates, idea implementation, and business impact—are tracked directly on the platform to ensure transparency and continuous improvement. Annual strategic themes guide the evolution of initiatives, allowing the organization to build innovation capabilities over time while adapting to feedback and shifting priorities.

Conclusions

This paper presents a data-driven framework for fostering innovation culture within large, multi-entity organizations. Based on empirical analysis, three core dimensions—*be inclusive*, *be specific*, and *provide fundamentals*—were identified as critical levers for enabling innovation at scale. These insights informed the design of targeted interventions, including Innovation Week, Hackathons, and a centralized innovation platform, each supported by measurable outcomes and implementation pathways. The proposed multi-year roadmap emphasizes leadership commitment, cross-functional collaboration, and continuous learning. Key performance indicators are tracked directly via the platform to ensure transparency and adaptive refinement.

The analysis demonstrates how organizational data and direct employee feedback can be used to diagnose cultural dynamics and prioritize interventions that build innovation capabilities. The evaluation approach as well as the framework is designed to be scalable and transferable across industries seeking to embed innovation as a core organizational function. Its success depends on broad participation and sustained commitment, reinforcing that innovation must be a collective effort supported by structure, strategy, and accountability with clear leadership commitment.

As the energy and petrochemical sectors continue to evolve, embedding innovation culture becomes a strategic imperative. While the findings of this pilot study across three multinational companies are promising, the current dataset is limited in scope. The authors therefore recommend conducting a broader,

cross-regional survey to capture cultural, organizational, and regulatory nuances more comprehensively. Future research should also explore the longitudinal impacts of such frameworks and assess their adaptability across different industry contexts. Ultimately, the transformation of innovation culture relies on inclusive engagement and the integration of innovation into everyday business practices.

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Nomenclature.

Abbreviations

AI	—	Artificial Intelligence
CCS	—	Carbon Capture and Storage
EDA	—	Exploratory Data Analysis
EB-1	—	Executive Board Level minus 1
HSSE	—	Health, Safety, Security, and Environment
IT	—	Information Technology
KPI	—	Key Performance Indicator
NPS	—	Net Promoter Score
PIMs	—	Performance Importance Matrices
R&D	—	Research and Development
TURF	—	Total Unduplicated Reach and Frequency

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